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Research Article

# Evaluation of Hepatoprotective Efficacy of Majoon-e-Dabeed-ul-ward Against Acetaminophen-Induced Liver Damage: A Unani Herbal Formulation

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| Strategy, Management and Health Policy | | | | |
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| Enabling Technology, Genomics, Proteomics | Preclinical Research | Preclinical Development Toxicology, Formulation Drug Delivery, Pharmacokinetics | Clinical Development Phases I-III Regulatory, Quality, Manufacturing | Postmarketing Phase IV |

**ABSTRACT** A Unani herbal formulation known as Majoon-e-Dabeed-ul-ward (MD) was evaluated for hepatoprotective effect against acetaminophen (APAP; 2 g/kg p.o.)-induced liver damage. The latter was evidenced by elevated levels of aspartate transaminase (AST), alanine transaminase (ALT), serum alkaline transaminase (SALP), lactate dehydrogenase (LDH), bilirubin, albumin, urea, and creatinine in experimental animals. Increased levels of lipid peroxidation were associated with a concomitant decline in reduced glutathione levels, adenosine triphosphatase (ATPase), and glucose-6-phosphatase (G-6-Pase) caused by APAP treatment. Treatment with MD (250,500, and 1,000 mg/kg p.o.) reverse the altered levels of AST, ALT, SALP, LDH, bilirubin, albumin, urea, and creatinine in a dose-dependent manner. Significant restoration was found in LPO, GSH content and metabolic enzymes (ATPase and G-6-pase) were seen after therapy. The herbal formulation, MD showed hepatoprotective efficacy against APAP-induced liver damage. Drug Dev Res 72:346–352, 2011. © 2010 Wiley-Liss, Inc.

**Key words:** hepatoprotective; Majoon-e-Dabeed-ul-ward; acetaminophen; liver disease

## INTRODUCTION

Liver disorders are a worldwide health problem. Liver is the organ responsible for drug metabolism and is a sensitive target site for substances undergoing biotransformation [Socha et al., 1992]. Liver diseases can be caused by toxic chemicals, excessive alcohol consumption, infections, and autoimmune disorders. The world market for natural products such as herbal, aromatic, and medicinal plants (HAMP) is expanding rapidly due to their phytomedicinal, neutraceutical, and phytocosmetic benefits [Roy and Das, 2010].

In India, numerous medicinal plants have been evaluated for the treatment of liver disorders in the traditional system of medicine, with some having been evaluated for their hepatoprotective actions against hepatotoxins. Currently available hepatoprotective herbal drugs are not sufficiently active to treat severe liver disorders; thus there is a need to develop more effective hepatoprotective drugs.

A single drug is unlikely to be effective for all types of severe liver diseases [Doughari et al., 2009]. An effective formulation using indigenous medicinal plants, with proper pharmacological experiments and clinical trials may thus provide a satisfactory alternative. Polyherbal therapy is a pharmacological

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principle that has the advantage of producing maximum therapeutic efficacy with minimum side effects and, according to the World Health Organization (WHO), 80% of the world population uses plant-based remedies as their primary form of health care [Sandhu et al., 2010]. The WHO is establishing definitive guidelines for clinical research methodologies and the appraisal of effectiveness of traditional medicines [Shahjahan et al., 2004; Shahani, 1999]. There is also an increasing interest in natural antioxidants (e.g., phenols), present in medicinal and dietary plants, that might help prevent oxidative damage. Natural antioxidants may be effective treatments for several pathologies not only as conventional hydrogen-donating compounds (antiradical activity) but, more importantly, may exert modulatory effects via actions on antioxidant, drug metabolizing and repairing enzymes as well as modulating as signaling molecules in important cascades for cell survival.

Increased mitochondrial stress and the production of reactive oxygen species (ROS) and reactive nitrogen species (RNS) have been implicated in APAP-induced hepatotoxicity [Ebong et al., 2008]. Numerous medicinal plants and their formulation are used for the treatment of liver disorders in ethnomedical practice and traditional system of medicine in India [Firenzuoli and Gori, 2007; James et al., 2003; Kumaran et al., 2009].

To develop a useful hepatoprotective herbal formulation, there is a need to evaluate the formulation for desired properties such as antioxidant and hepatoprotective efficacy. Majoon-e-Dabeed-ul-ward (MD) is a Unani polyherbal formulation with and official monograph in National Formulary of Unani Medicine (NFUM), prepared by mixing parts of 21 medicinal plants and used in Indian System of Medicine (Table 1). MD contains flavonoids, glycosides such as crocin, present in *Crocus sativa* [Chen et al., 2008]; gentian present in *Gentian olivieri* [Singh, 2008]; cascutin present in *Cusseta reflexa* [Sharma et al., 2010]; querecetin and kaempferol, present in *Rosa damascene* [Schiieber et al., 2005]; esculetin, present in *Cichorium intybus* [Gilani et al., 1998]; tannins, terpenes, steroids, and saponins, present in *Asarum europaeum* [Saifuddin et al., 2010]; *Commiphora opobalsamum* [Al-Howiriny et al., 2004] and *Aquilaria agallocha* [Dash et al., 2008], myrcene, limonene, terpinen-4-ol, and α-pinene, present in *Pistacia lentiscus* [Castola et al., 2000]; sesquiterpenes such as lignans and neolignans, present in *Nardostachys jatamansi* [Mi-Young et al., 2010]; and cinnamaldehyde, present in *Cinnamomum zeylanicum* [Walunj et al., 2010]. Flavonoids are phenolic compounds that exert multiple biological effects [Hazra et al., 2008]. In the present study hepatoprotective effect of MD was evaluated against APAP induced hepatic damage.

## Animals

Adult female albino Sprague-Dawley rats (12–14 weeks old, $160 \pm 10$ g body weight) were randomly selected from a departmental animal facility where they were housed in polypropylene cages under

**TABLE 1. Medicinal Plant Ingredients of Majoon-e-Dabeed-ul-ward, a Unani Polyherbal Formulation**

| Unani name | Botanical name | Family name | Part used | Weight (g) |
|---|---|---|---|---|
| Sumbul-ut-Teeb | *Nardostachys* | Valerianaceae | Whole plant | 10 |
| Mastagi | *Pistacia lentiscus* | Anacardiaceae | Resin | 10 |
| Zafran | *Crocus sativus* | Crocoideae | Flower (stigma) | 10 |
| Tabasheer | *Bambusa bambos* | Poaceae | Resin | 10 |
| Darchini | *Cinnamomum zeylanicum* | Lauraceae | Stem bark | 10 |
| Izkhar | *Cymbopogon jwarancusa* | Poaceae | Whole plant | 10 |
| Asaroon | *Asarum europaeum* | Aristolochiaceae | Root | 10 |
| Qust Shireen | *Saussurea hypoleuca* | Asteraceae | Root | 10 |
| Gul-e-Ghafis | *Gentiana olivieri* | Gentianeae | Root | 10 |
| Tukm-e-kasoos | *Cuscuta reflexa* | Cuscutaceae | Seed | 10 |
| Majeeth | *Rubia cordifolia* | Rubiaceae | Root | 10 |
| Luk Maghsool | *Coccus lacca* | Kerriidae (Lac insect family) | Secretion by leaves, insect | 10 |
| Tukm-e-Kasni | *Cichorium intybus* | Asteraceae | Seed | 10 |
| Tukm-e-Karafs | *Apium graveolens* | Apiaceae | Seed | 10 |
| Zarawan Taweel | *Aristolochia donga* | Asclepiadaceae | Root | 10 |
| Habb-e-Balsan | *Commiphora opobalsamum* | Burseraceae | Fruit | 10 |
| Ood-e-Hindi | *Aquilaria agalocha* | Thymeleceae | Stem | 10 |
| Qaranful | Syzygium aromaticum | Myrtaceae | Dried buds | 10 |
| Heel Khurd | *Elettaria cardamomum* | Zingiberaceae | Dried fruit | 10 |
| Waaraq-e-Gul-e-Surkh | *Rosa damascena* | Rosaceae | Petals | 200 |
| Qand Safed (granular sugar) | | | A product made from sugar cane | 600 |

uniform husbandry conditions of light (12-h light–dark cycle at 30°C) and dark (10 h) with a temperature of $25 \pm 2$°C and relative humidity of 60–70%. Animals were provided with pellet diets (Pranav Agro Industries, New Delhi, India) and water ad libitum. Experimental protocols were approved by an Institutional Ethical Committee (CPCSEA/50/01/01/A), Jiwaji University, Gwalior (India).

### Drug and Chemicals

MD was obtained by Central Council for Research of Unani Medicine (CCRUM), New Delhi. Silymarin (Sigma Aldrich Chemicals Pvt, India) and acetaminophen (APAP) from GlaxoSmithKline (Batch 0103). All other reagents used in the study were of analytical grade.

### Experimental Design

APAP was suspended in warm distilled water and administered (2 g/kg p.o., once only) according to Nirala and Bhadauria [2008]. MD at doses of 250, 500, and 1,000 mg/kg p.o. and silymarin (50 mg/kg p.o.) as a positive control [Anand et al., 1997] were administered to the animals after 24 h of APAP treatment. Rats were divided into seven groups of six animals each and treated as follows:

Group 1: distilled water, control.
Group 2: MD (1,000 mg/kg p.o.) treatment control.
Group 3: APAP (2 g/kg p.o.) once experimental control.
Groups 4–6: APAP (as in group 3)+MD (250, 500, and 1,000 mg/kg p.o., respectively).
Group 7: APAP (as in group 3)+silymarin (50 mg/kg p.o.).

Animals were sacrificed 24 h after the last treatments; blood was collected and allowed to clot, and serum was separated at 2,000 rpm for 20 min and stored at −20°C in a deep freezer. Various blood and tissue biochemical parameters were performed in serum and liver tissues.

### Biochemical Studies

Various biochemical parameters were measured, including serum aspartate transaminase (AST) and alanine transaminase (ALT) [Reitman and Frankel, 1957] serum alkaline phosphatase (SALP) [Fiske and Subbarow, 1925], lactate dehydrogenase (LDH) (no. 11760200011730) and albumin (no. 118275.0001), bilirubin (no. 103333.0001), urea (no. 117618.0001), creatinine (no. 103385.0001) (Merck kit method, as per directions).

### Tissue Biochemical Assay

GSH was measured using dithionitrobenzoic (DTNB) acid [Brehe and Burch, 1976]. Optical density was recorded immediately at 412 nm. The GSH level was calculated using an extinction coefficient of $13{,}600\,\mathrm{M\,cm^{-1}}$ and expressed as µmol GSH/g tissue. Malondialdehyde (MDA), a frequently used indicator of lipid peroxidation, was assayed for lipid peroxidation (LPO) [Sharma and Krishnamurthy, 1968], and LPO was expressed in terms of nmol MDA/g tissue, using an extinction coefficient of $1.56 \times 105\ \mathrm{M^{-1}\,cm^{-1}}$. Glucose-6-phosphatase [Baginski et al., 1974] and ATPase [Seth and Tangari, 1966] were measured in liver homogenate.

### Histological Observations

For light microscopic observations, liver samples were fixed in Bouin's fixative and processed routinely for embedding in paraffin. Tissue sections of 5-µm thickness were stained with hematoxylin and eosin (H&E) and examined under a photomicroscope.

### Statistical Analysis

The data for various biochemical parameters were analyzed by one-way analysis of variance (ANOVA), followed by Duncan's multiple range test ($P \leq 0.05$) was considered statistically significant [Snedecor and Cochran, 1989].

## RESULTS

### Effect of MD on Hepatorenal Markers for APAP-Induced Liver Damage

Table 2 shows that the levels of the hepatic enzymes AST, ALT, LDH, and SALP were increased in blood after APAP treatment ($P \leq 0.05$). Treatment with MD at all doses (250, 500, and 1,000 mg/kg) normalized enzyme activities ($P \leq 0.05$). The 1,000-mg/kg dose of MD showed significant reversal of the effects of APAP as with the control group. MD (1,000 mg/kg) had no effect on enzymatic activity. Table 3 shows a sharp elevation in albumin, bilirubin, urea, and creatinine content at 24 h after APAP administration ($P \leq 0.05$). MD at all doses examined showed regaining of the above parameters. At the 1,000-mg/kg dose, reversal of the APAP effects was 64–87% (one way ANOVA ($P \leq 0.05$)).

### Effect of MD on Oxidative Markers (LPO and GSH) Induced by APAP

Table 3 shows the altered level of oxidative markers in response to APAP treatment. APAP increased MDA levels and decreased GSH content as compared with controls ($P \leq 0.05$). Treatment with MD reversed altered level of oxidative markers in a dose-dependent manner and showed better maximum protection at the 1,000-mg/kg dose in LPO (86.4%) and GSH content (90.9%), to the same extent as the positive control silymarin ($P \leq 0.05$).

**TABLE 2. Effect of Majoon-e-Dabeed-ul-ward Against Acetaminophen-Induced Alterations in Liver Function Tests**

| Treatment | AST (IU/L) | ALT (IU/L) | LDH (U/L) | SALP (mg/Pi/h/100 ml) |
|---|---|---|---|---|
| Control | 64.0±5.85 | 47.0±4.11 | 42.0±3.93 | 200±19.3 |
| MD, 1,000 mg/kg | 69.0±5.09 | 50.0±3.16 | 40.0±4.34 | 202±15.0 |
| APAP | 270±24.3† | 328±18.23† | 155±10.3† | 550±41.5† |
| APAP+MD 250 mg/kg | 145±10.0* (60.6%) | 170±10.0* (56.2%) | 70.0±6.00* (75.2%) | 280±18.5* (77.1%) |
| APAP+MD 500 mg/kg | 120±11.6* (72.8%) | 150±11.7* (62.2%) | 65.0±4.89* (79.6%) | 250±12.3* (85.7%) |
| APAP+MD 1,000 mg/kg | 98.2±11.6* (83.3%) | 130±11.6* (70.4%) | 60.0±4.24* (84.0%) | 210±12.3* (97.1%) |
| APAP+S 50 mg/kg | 70.0±8.24* (97.2%) | 110±8.24* (77.5%) | 50.0±3.16* (92.9%) | 205±20.3* (98.6%) |
| F-value (at 5% level) | 39.0[a] | 73.5[a] | 46.0[a] | 40.4[a] |

APAP, acetaminophen; MD, Majoon-e-Dabeed-ul-ward; S, silymarin; %, percentage protection. Values are mean±SE; $N=6$.
†$P\leq0.05$ vs control; *$P\leq0.05$ vs APAP.
[a]ANOVA = significant.

**TABLE 3. Effect of Majoon-e-Dabeed-ul-ward Against Acetaminophen Intoxication**

| Treatments | Albumin (g/dl) | Bilirubin (mg/dl) | Urea (mg/dl) | Creatinine (mg/dl) |
|---|---|---|---|---|
| Control | 3.20±0.28 | 0.20±0.08 | 19.0±1.32 | 0.25±0.03 |
| MD, 1,000 mg/kg | 3.20±0.29 | 0.21±0.04 | 19.1±1.39 | 0.20±0.02 |
| APAP | 4.80±0.43 | 1.30±0.29† | 50.0±3.16† | 3.10±0.08† |
| APAP+MD 250 mg/kg | 4.10±0.35 (43.7%) | 1.10±0.17 (18.1%) | 45.0±3.74 (16.1%) | 2.00±0.84 (38.5%) |
| APAP+MD 500 mg/kg | 3.60±0.35 (75.0%) | 0.90±0.11 (36.3%) | 40.0±3.16 (32.2%) | 1.50±0.05* (56.1%) |
| APAP+MD 1,000 mg/kg | 3.40±0.48 (87.5%) | 0.40±0.06* (81.8%) | 30.0±2.69* (64.5%) | 0.60±0.02* (87.7%) |
| APAP+S 50 mg /kg | 3.30±0.30 (93.7%) | 0.22±0.05* (98.1%) | 20.0±1.46* (96.7%) | 0.40±0.04* (94.7%) |
| F-value (at 5% level) | 1.70[ns] | 13.4[a] | 32.4[a] | 29.0[a] |

APAP, acetaminophen; MD, Majoon-e-Dabeed-ul-ward; S, silymarin; %, percentage protection. Values are mean±SE; $N=6$.
†$P\leq0.05$ vs control; *$P\leq0.05$ vs APAP.
[a]ANOVA = significant; ns, not significant.

**TABLE 4. Effect of Majoon-e-Dabeed-ul-ward on Acetaminophen-Induced Alterations in Tissue Biochemistry**

| Treatments | LPO (nmoles MDA/mg protein) | GSH (μmole/g) | ATPase (mg Pi/100 ml/min) | G-6-Pase (μmole Pi/min/g liver) |
|---|---|---|---|---|
| Control | 0.32±0.02 | 8.20±0.67 | 1998±100 | 6.30±0.59 |
| MD, 1,000 mg/kg | 0.30±0.02 | 8.00±0.84 | 2000±113 | 6.29±0.52 |
| APAP | 1.80±0.17† | 3.80±0.22† | 677.0±47.3† | 3.00±0.28† |
| APAP+MD 250 mg/kg | 0.82±0.06* (66.2%) | 7.20±0.65* (77.2%) | 1207±69.8* (40.1%) | 4.53±0.44* (46.3%) |
| APAP+MD 500 mg/kg | 0.75±0.18* (70.9%) | 7.40±0.45* (81.8%) | 1407±76.4* (55.2%) | 4.90±0.66* (57.5%) |
| APAP+MD 1,000 mg/kg | 0.52±0.11* (86.4%) | 7.80±0.59* (90.9%) | 1880±116* (91.0%) | 5.86±0.69* (86.6%) |
| APAP+S 50 mg/kg | 0.42±0.03* (93.2%) | 8.00±0.48* (95.4%) | 1899±107* (92.5%) | 6.29±0.48* (99.6%) |
| F-value (at 5% level | 42.1[a] | 8.11[a] | 34.0[a] | 63.4[a] |

APAP, acetaminophen; MD, Majoon-e-Dabeed-ul-ward; S, silymarin; %, percentage protection. Values are mean±SE; $N=6$.
†$P\leq0.05$ vs control; *$P\leq0.05$ vs APAP.
[a]ANOVA = significant.

## Effect of MD on ATPase and G-6-Pase

Table 4 shows significant decreases in the activities of ATPase and G-6-Pase after APAP exposure ($P\leq0.05$). Administration of MD (250, 500, and 1,000 mg/kg) significantly reversed these effects with maximal recovery of ATPase (91%) and G-6-Pase (86.6%) activity at an MD dose of 1,000 mg/kg p.o. at 5% level.

## Histopathological Observations

Figure 1 shows the histology of liver after different treatments. Light microscopic evaluation showed regular morphology of the liver parenchyma with well-designed hepatic cells and sinusoids in controls (Fig. 1A). APAP treatment produced severe liver injury with prominent congestion in sinusoidal spaces and high accumulation of inflammatory cells and nuclear disintegration and chromatolysis with patchy areas of fatty degeneration (Fig. 1B). The groups treated with 250 mg/kg (Fig. 1C) and 500 mg/kg dose (Fig. 1D) of MD demonstrated less hepatocyte necrosis and fatty changes in the liver tissues. Treatment with MD (1,000 mg/kg, Fig. 1E) significantly

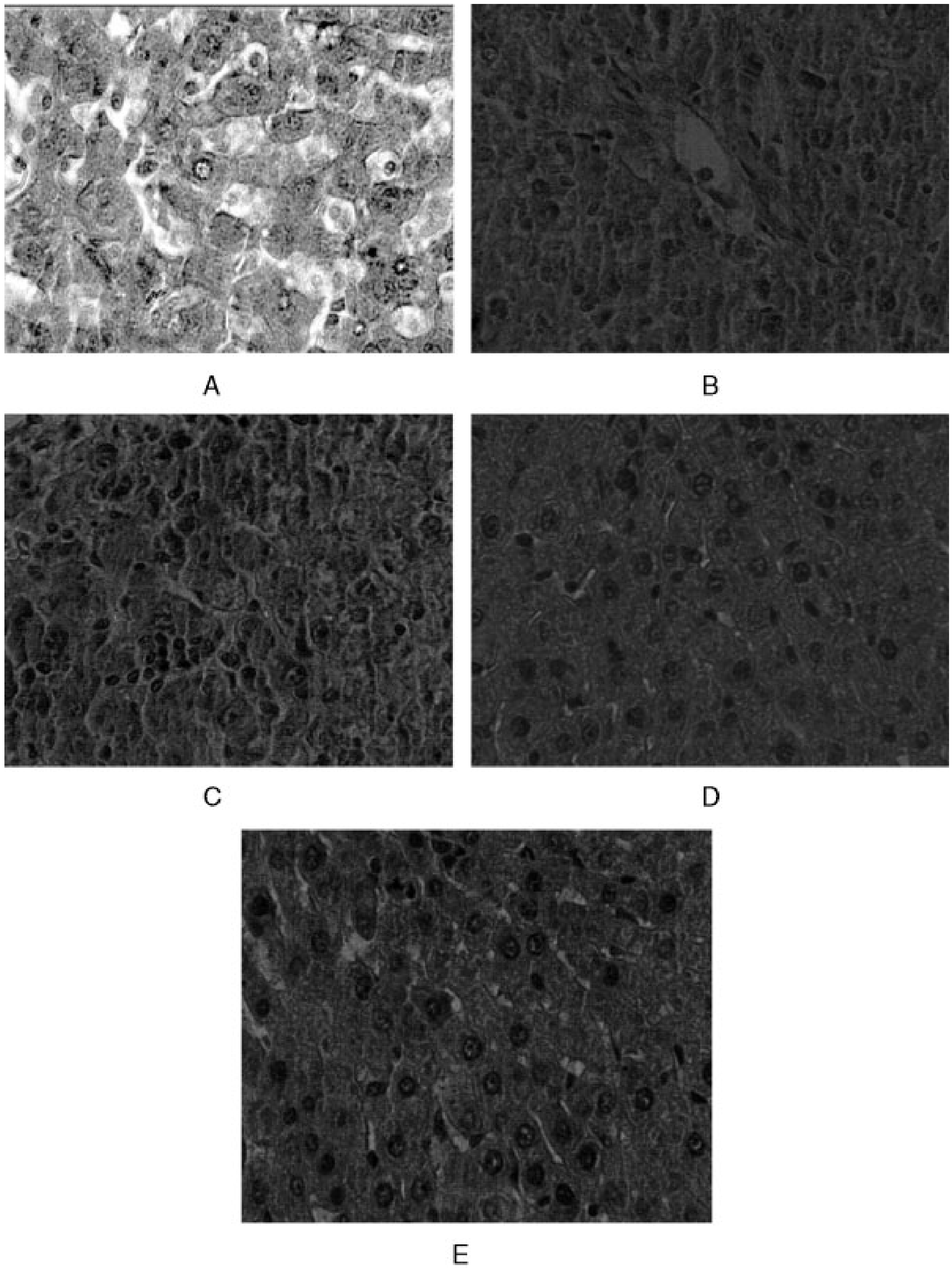


**Fig. 1.** Representative photomicrographs of histological changes showing effect of the MD on the rats intoxicated with APAP. $\times 400$. **A**: Normal. **B**: APAP experimental control. **C**: APAP+MD (250 mg/kg). **D**: APAP+MD (500 mg/kg). **E**: APAP+MD (1,000 mg/kg). All $\times 400$. (400X).

attenuated the necrotic changes and cellular architecture of the liver.

## DISCUSSION

The present study evaluated the hepatoprotective efficacy of MD against APAP-induced liver toxicity in experimental model. APAP is normally eliminated mainly as sulfate and glucoronide with only 5% of the acetaminophen converted into *N*-acetyl-*p*-benzoquineimine (NAPQI). Administration of toxic doses of APAP saturates sulfation and glucuronidation pathways, resulting in a higher percentage of APAP being oxidized to highly reactive NAPQI by cytochrome P-450 enzymes. Semiquinone radicals, obtained by the one-electron reduction of NAPQI, covalently bind to macromolecules of the cell membrane and enhance lipid peroxidation. Higher doses of APAP can alkylate and oxidize intracellular GSH and protein thiol groups resulting in depletion of liver GSH pools leading to increased lipid peroxidation and liver damage [Maheswari et al., 2008; Marotta et al., 2009].

In liver injury, the transport function of the hepatocytes is altered, especially the permeability of plasma membrane, leading to increased enzyme levels in serum yielding hepatorenal markers indicative of liver injury [Felipe et al., 2008; Vadivu et al., 2008]. Thus, there is a direct correlation between increases in serum enzyme activity and severity of damage to liver indicating necrosis caused by APAP administration. The stabilization of hepatorenal markers by MD therapy improves the functional status of liver cells.

The products of lipid peroxidation are themselves reactive species and lead to extensive membrane and cellular damage [Kapur et al., 1994; Cotran et al., 1999]. These free radicals trigger cell damage via two mechanisms, namely covalent binding to cellular macromolecules and lipid peroxidation, which affect the ionic permeability of the membrane, altering the fluidity of the membrane structure. Reduced glutathione (GSH) is an important endogenous antioxidant system found in particularly high concentrations in liver and has a key protective function. Excessive production of free radicals results in oxidative stress, leading to macromolecule damage. GSH was also depleted when lipid peroxidation is enhanced after APAP treatment indicating significant hepatic damage. MD reverses the reduced GSH and reduces MDA levels to prevent tissue injuries. The standard hepatoprotective drug, silymarin maintained the decreased lipid peroxidation and GSH level to the normal limits in the liver indicating improved metabolic activity and cellular stability. Damage to the cellular membrane from lipid peroxidation also leads to decreased activity of the endoplasmic reticulum membrane-bound enzymes such as G-6-Pase. Our studies showed the acute exposure to APAP in rats significantly decreased ATPase and G-6-pase activities in liver which might be due to the membrane fragility and permeability of the organs.

Therapy for MD at all doses significantly restored metabolic enzyme activity, indicating improved physiological function in liver tissue. APAP administration elicited extensive changes in liver cells, including marked fatty degeneration, obvious collagen deposition, hepatocyte ballooning, and infiltration of inflammatory cells in liver interstitium. MD protection against APAP induced biochemical and histopathological changes in liver.

In conclusion, MD was found to demonstrate hepatoprotective efficacy against APAP-induced hepatic damage. It is postulated that its antioxidant properties may be involved in the hepatoprotective activity.

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